



OpenStreetMap Uganda

90 Kanjokya Street

Kampala, Uganda

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MAPPING MANUFACTURERS' CAPABILITIES IN UGANDA

Authors:- **OpenStreetMap Uganda**
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Executive Summary

OpenStreetMap Uganda team mapped Manufacturers capabilities around Kampala industrial Area, Namanve Industrial Park and other areas around Kampala and Greater Kampala. During this data collection exercise we targeted large, medium and small scale manufacturing facilities in order to boost local manufacturing and supply. We were able to collect data from over **400** manufacturers.

Project Objectives

The project was aimed at mapping out different manufacturers across Kampala and Greater kampala to determine the capabilities of these manufacturers based on the products and number of products they manufacture per day and their inputs.

- Map out the name, address and location of the manufacturing facility
- Capture information on the products they manufacture and their daily production capacity
- Capture information on the utilities a manufacturing facility utilizes
- Capture information on the required inputs for the production process of the manufacturer

Methodology

During this data collection process we employed the methodology below to achieve our intended objectives.

- **Planning and Literature Review**

During this process a plan to carry out this data collection was developed. The plan included which platform to use for the data collection, number of teams needed to carry out this data collection and the required period. Review of literature from the previous manufacturers mapping done by Field Ready with support of Humanitarian OpenStreetMap Team (HOT) to help better prepare for this project and to anticipate the different challenges and how to solve them.



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● Form Design

The survey form was designed using the [Kobotoolbox](#) and excel spreadsheet software. All the basic questions were included as required by the project to ensure complete data in reference to the [OpenKnowWhere Standards](#).

The major questions captured were the name of the Facility, Address, Contact, Location, Production Capacity and Required inputs.

● Form Review

Form review and testing was done internally together with the technical team from the Internet of Production Alliance. During this review questions that were missed were added and others which needed some technicalities were reorganized so as to meet the high standard of data required.

● Enumerators

A call was put out to hire volunteers who would then carry out the manufacturers' mapping. Over 20 interested parties applied and were vetted to remain with five candidates selected. Two (2) Ladies and three (3) gentlemen.

● Mobile Data Collection Tool

The mobile data collection tool to use for this data collection was [KoboCollect](#) on Android. KoboCollect was selected because of the team's vast experience using it and that it would give the best possible results compared to other tools.

● Data Enumerators Training

After the section of Data Enumerators and Data Collection Tool, we held a training for one day for the enumerators. During this process we had an introduction to this data collection and objectives. We discussed and went through the data collection tool to understand how each question is supposed to be answered and the data collection ethics.

After this we went through the application installation of the KoboCollect on the different smartphones given to the data collectors. They were then configured to access the form and the form was downloaded onto each of their devices. The enumerators then were taught how to use the form on the mobile devices, how to save the form and how to send the form to the KoboToolBox.

Data Collection

After the training the data collectors then went out to collect data for 8 days under the supervision of one of our mapping supervisors



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Results

The team was able to map a total of over **826** Machines through this data collection period of 8 days.

Facility Types Mapped

During this data collection process the team was mapping out different types of manufacturing facilities around kampala and greater kampala and the summary of these facilities is as in the table below;

Facility Type	Number of Facilities
Private	135
Government	1
Workshop	208

Table 1:- Summary of Facility types Mapped

Materials Input Mapped

The materials input for the different production processes from the different manufacturing facilities were as follows below;

Materials	Number
Metals	181
Wood	103
Elastomers	2
Electronics	1
Plastic	25
Ceramics	7
Others	157

Table 2:- Summary of Materials Input Mapped



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Challenges Faced

- Some manufacturing facilities wanted to keep their information private and so they could not share.
- Due to the rainy season, The data collection activity was partially affected, delaying data collectors to go to the field.
- Some of the information needed to be obtained from administrators of the facility hence leading to appointments with them before getting the information.
- Most of the facilities are far away from each other making the enumerators tiresome at the end of the day after long travels.

Recommendation

- For the next data collection, the lead team should get in touch with manufacturers associations, organizations or unions to assist them get information from manufacturers belonging to their groups.
- Information deemed sensitive like machines owned by manufacturers should be reconsidered during the next data collection phase.
- Target more of the small scale manufacturers as they are more willing to share their information

Conclusion

During a period of 10 days and a team of 7 members, We were able to collect data from different manufacturers around kampala and greater kampala. The data is available on the KoboToolBox and we shall be assembling a team to clean this data and upload it to the OpenStreetMap Platform. The data will also be shared with IOPA to clean and put up the data on a platform of their own choice.

Overall, this data collection process was very successful.



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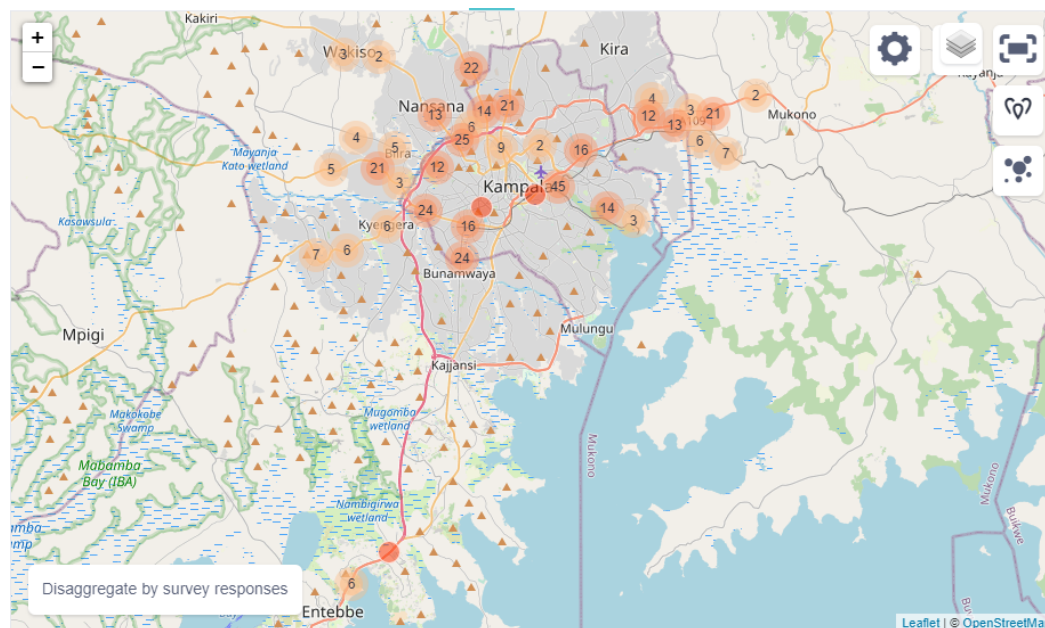
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Appendix

Online Map



Map Showing Distribution of Facilities Mapped

Facility Type
☐ Workshop
☐ Government
☐ Private
Affiliation
☐ Individual Clients
☐ Organisations
Name of Facility Partner, Funder or Organisation?
Does your company have any ISO certifications?
What is the status of your facility?

Image of the form as seen in Kobo



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A	B	C	D	E
1 type	name	label	hint	required
2 start	start			
3 end	end			
4 select_one s18zi25	enumerator	Enumerator		false
5 select_one mg1yb16	does_the_facility_agree_to_participate	Does the Facility Agree to Participate?		false
6 text	facility_name	Facility Name		false
7 select_one gg23k22	country	Country		false
8 geopoint	record_location_of_facility	Record Location of Facility	Please make sure you stand under openskies without any obstruction	false
9 text	owner_of_the_facility	Owner of the Facility		false
10 text	contact_information	Contact Information		false
11 text	email	Email		false
12 select_one bf9k02	facility_type	Facility Type		false
13 select_one bj0ui10	affiliation	Affiliation		false
14 text	facility_partner_funder_or_org	Name of Facility Partner, Funder or Organisation?		false
15 text	any_certifications	Does your company have any ISO certifications?		false
16 select_one uz5gf26	facility_status	What is the status of your facility?		false
17 time	facility_opens_at	What time does your facility open?		false
18 time	facility_closes_at	What time does your facility close?		false
19 select_multiple xe65z01	working_days	How many days in a week do you work?		false
20 date	date_founded	When was your company founded?		false
21 text	maintenance_schedule	How often do you perform maintenance to your equipment or facility?		false
22 select_one storage	storage_capacity	Whats the storage capacity of your facility?	In cubic meters	false
23 select_one floor	floor_size	What is the estimate of the floor size of facility?	In square meters	false
24 select_one vi0vy90	loading_dock	Does your facility have a loading dock?		false
25 select_one mz3tz63	back_up_generator	Do you have a backup generator?		false
26 select_one gt4uz04	uninterrupted_power_supply	Does your facility have uninterrupted power supply?		false
27 select_one yb8rd61	facility_access_type	Whats the access type of your facility?		false
28 select_one nx0ma08	road_access	Does your facility have a road access?		false
29 select_one nt85v42	wheel_chair_access	Does your facility have a wheel chair access?		false
30 integer	number_of_people_working	How many people are working in your facility?		false
31 text	name_of_the_person_in_charge_of_the_manufacturing? (Maker, Craftsman, builder)			false

Excel design of the form

KoBoToolbox		OKW Data Collection2		408 submissions	
NEW		SUMMARY		FORM	
Table		DATA		SETTINGS	
Deployed 5		hide fields			
Draft 0		1 - 30 results			
Archived 0		Validation			
Reports		Show All			
Gallery		start			
Downloads		end			
Map		Enumerator			
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A picture of one of the industries we mapped



One of the field mappers in action



OSM MAPUGANDA-MOTIV DATA COLLECTION EXERCISE REPORT



Facility Type, Affiliation, Facility

✓

☐

(Blank)

✓

☐

Government

✓

☐

Private

✓

☐

Workshop

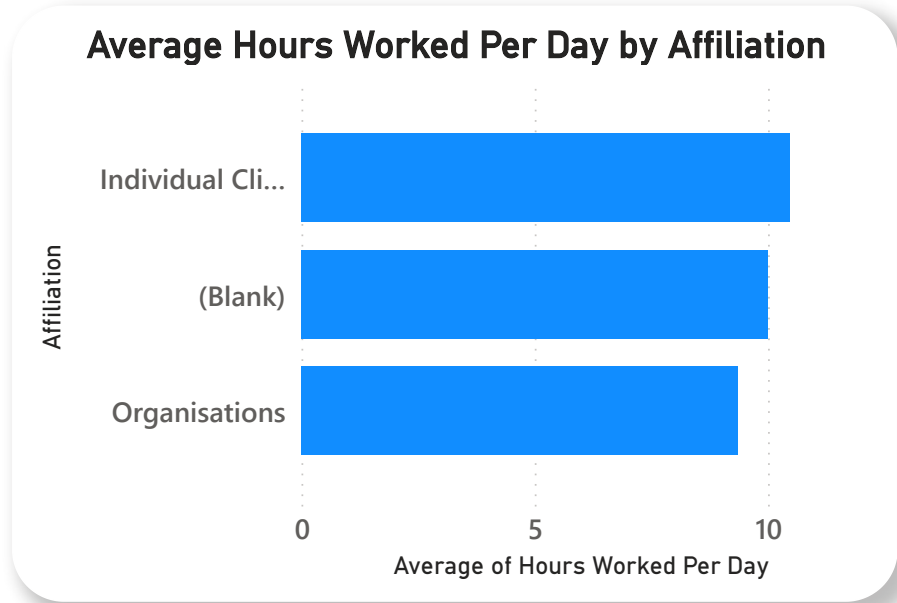
108.45

Average Number of Employees

Year of Founding

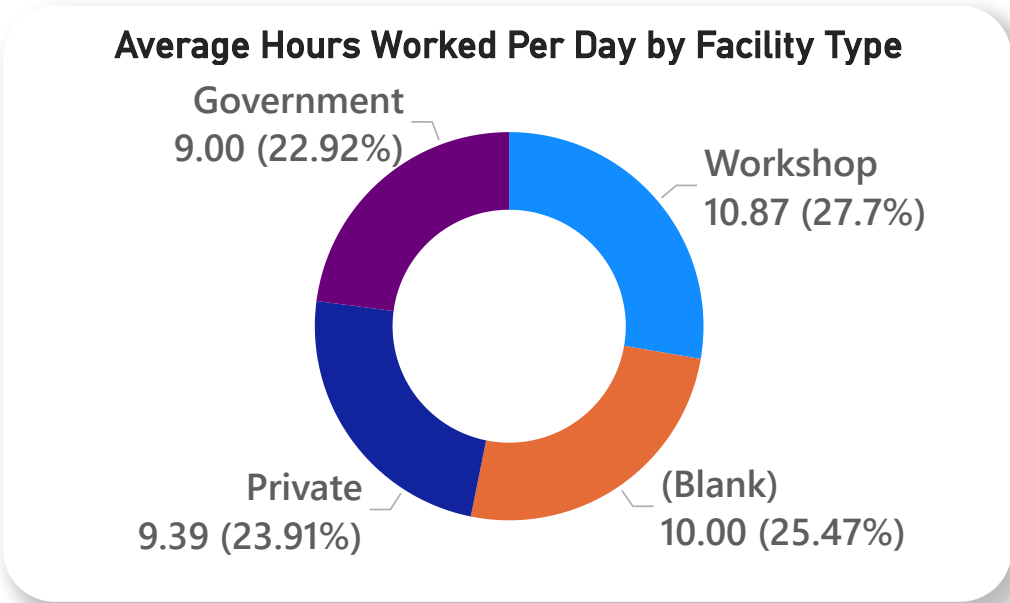
1910

2022



277

Total Number of Facilities

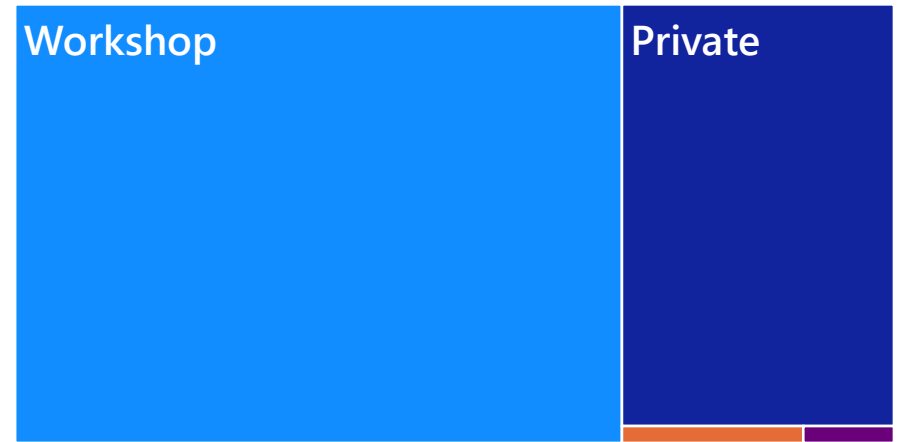




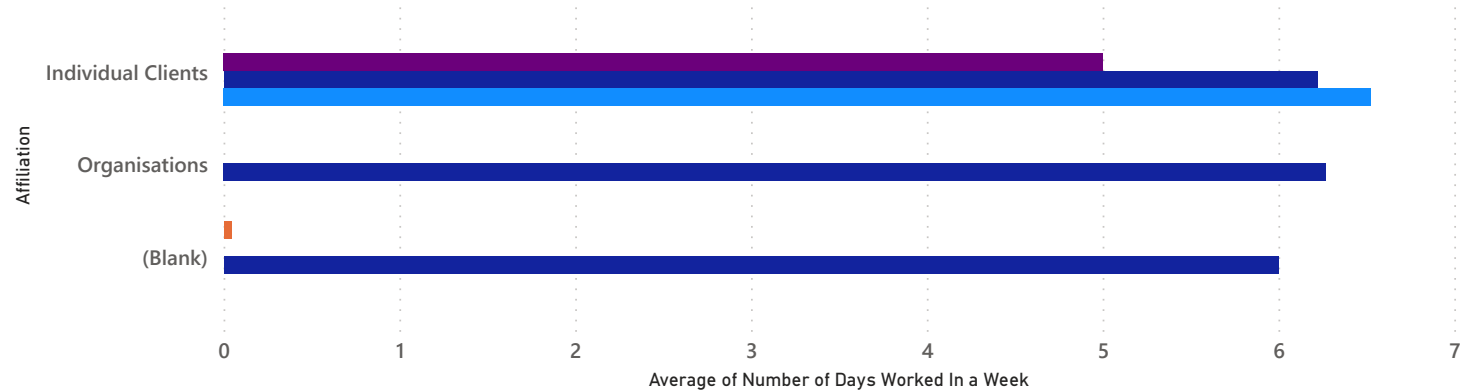
%GT Count of Machines by Origin

Origin of Machine	Percentage
Japan	36.58%
England	31.16%
China	25.49%
India	2.83%
Nil	1.72%
Total	100.00%

%GT Count of ISO Certifications by Facility Type

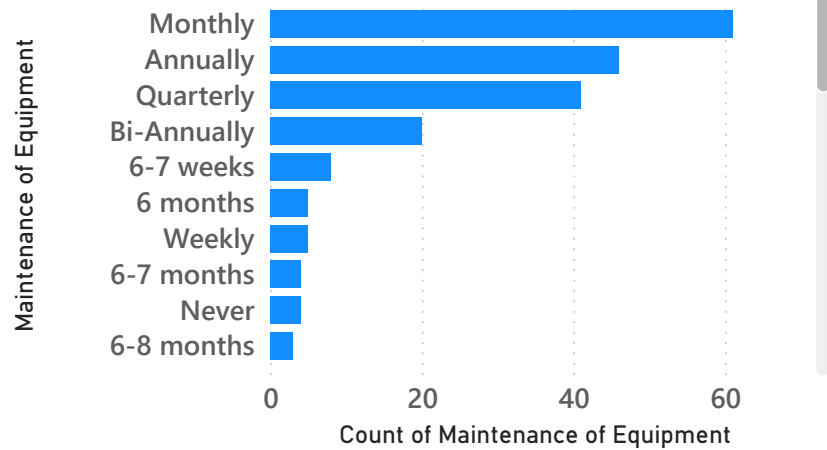


Average Number of Days Worked In a Week by Affiliation and Facility Type

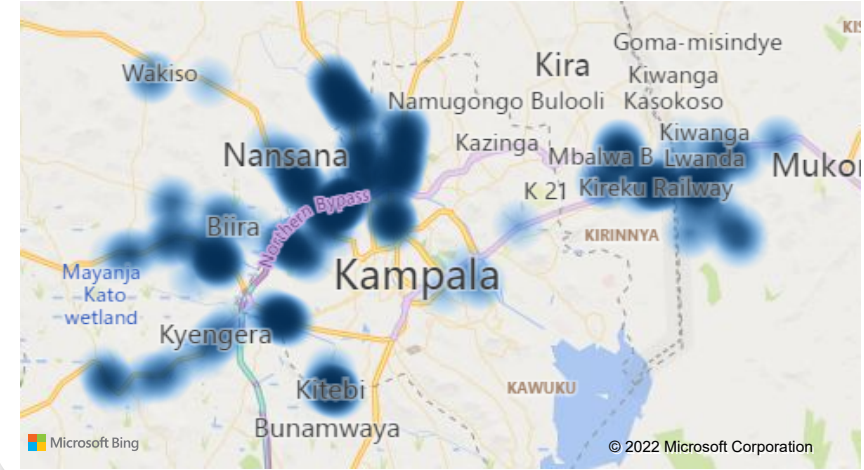




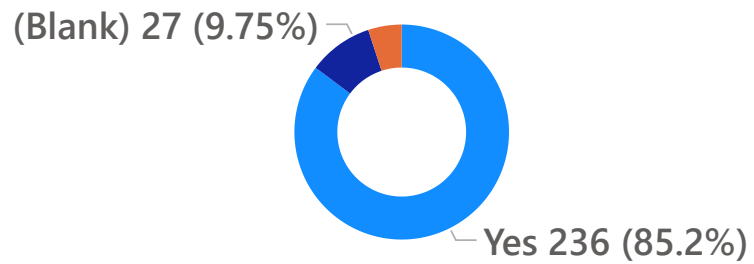
Count of Maintenance of Equipment by Maintenance of Equipment



Heatmap showing concentrations of facilities in study area



Facilities with Loading Docks



Facilities with Backup Generators

